

The Natural Step – Team #2

XTSE: NTR

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Company Overview

01 FOOTPRINT

- Headquartered in Saskatoon, Saskatchewan
- Production operations in Canada, the U.S. and Trinidad
- Headcount: 25,100

02 WHAT NUTRIEN DOES

- Produces and distributes essential crop nutrients
- Supplies growers with crop inputs and services
- Links large-scale production with the Nutrien Ag Solutions retail network

03 INDUSTRY

- Agricultural inputs and services
- Fertilizer mining and manufacturing
- Crop-input retail and distribution

04 PRODUCTS & SERVICES

- Potash, nitrogen and phosphate fertilizers
- Seed, crop protection and nutritional products
- Agronomy, digital tools and grower services

Why Nutrien?

Industry Leaders

NATURAL SYSTEMS

Water • land • minerals

HUMAN + ECONOMIC VALUE

Food security • livelihoods • long-term competitiveness

01 DIRECT TNS FIT

Heavy reliance on water, land, minerals. Environmental impacts tangible and assessable.

02 CLEAR SUSTAINABILITY

Fertilizer supports food security while creating pressure through emissions, water use, runoff and resource depletion.

03 STRONG SDG ALIGNMENT

Direct links to SDGs 2, 6, 7, 12, 13 and 15.

04 INTEGRATED SYSTEMS

The same issue can be tested through environmental, human and economic lenses revealing both value and trade-offs.

05 MEASURABLE ESG PERFORMANCE

Published targets and performance metrics allow evidence-based evaluation of whether actions match environmental impacts.

06 BUSINESS RESILIENCE

Dependence on water, energy and agricultural land connects sustainability challenges to long-term performance and competitiveness.

Stakeholders, Competitors, and Value Chain

Stakeholders

- Investors
- Farmers/Producers
- Growers/Customers
- Communities
- Suppliers
- Indigenous Peoples
- Employees
- Government/Regulators

Competitors

- Mosaic
  - Potash & Phosphate Fertilizers
- CF Industries
  - Nitrogen Fertilizers
- K+S Potash
  - Potash

Value Chain

- Upstream
  - Mining & Production
- Midstream
  - Logistics & Distribution
- Downstream
  - Products & Services

Key Competitors

Mosaic

K+S Potash Canada

CF Industries, Inc.

News & Notable Advancements

COMMUNITY & PROCUREMENT



$1.1B

potash procurement spend across 2020–2024



$567M

of that spend with Indigenous-owned companies



35%

of 2024 community giving directed to sustainable agriculture

49%

POTASH AUTOMATION

of potash ore tonnes were mined by automation in 2025 — our roadmap pushes this past 65%.

98%

NITROGEN RELIABILITY

record ammonia operating rate in H1 2025; debottlenecks completed at Redwater and Geismar.

<$60/t

COST OF PRODUCTION

controllable cash cost per potash tonne held below $60 in Q1 2026 — automation cuts cost, not just emissions.

NH_3

LOW-CARBON AMMONIA

EXMAR collaboration to build an ammonia-fuelled vessel positions ammonia as marine fuel, not just fertilizer.

Momentum is real — but efficiency-shaped. Automation, reliability and cost are advancing fastest; the absolute-emissions and worker-safety gaps in our SLCA remain open.

The Natural Step: Four System Conditions

A sustainable society does not systematically increase or degrade these four things.

SC1
Earth's Crust

Does not increase concentrations of substances extracted from the earth's crust (fossil fuels, metals, minerals).

SC2
Society-Made Substances

Does not increase concentrations of substances produced by society (synthetic chemicals, persistent compounds, N_2O).

SC3
Nature's Cycles

Does not degrade nature through overharvesting, land disturbance, or ecosystem/biodiversity loss.

SC4
Human Needs

Does not create structural barriers to people meeting their basic needs (safety, livelihood, community wellbeing).

OUR STOPLIGHT METHOD

GREEN = industry leader on this condition • **YELLOW** = industry average • **RED** = industry laggard, with citations behind every rating.

SLCA Stoplight Matrix

	Raw Materials	Production	Packaging & Distribution	Use	End of Life
SC1 · Materials from the Earth's crust	AVERAGE	LAGGARD	AVERAGE	LEADER	LAGGARD
SC2 · Man-made materials	AVERAGE	AVERAGE	AVERAGE	LAGGARD	AVERAGE
SC3 · Degradation of the biosphere	AVERAGE	AVERAGE	AVERAGE	LAGGARD	AVERAGE
SC4 · Undermining human needs	LEADER	LEADER	LEADER	AVERAGE	LEADER

Industry leader Industry average Industry laggard

FOUR RED CELLS — OUR HOTSPOTS

- SC1 · Production — fossil gas is the feedstock for ammonia, not just the fuel.
- SC1 · End of Life — tailings and salt stockpiles left on the landscape.
- SC2 · Use — nutrient runoff from applied fertilizer.
- SC3 · Use — biosphere degradation from over-application.

Sources: Nutrien 2024 Global Sustainability Report; Form 40-F (FY2024); team analysis.

Hot Spot 1 Permanent Contamination Risk: System Condition 1 at End of Life

Mining Waste with No Viable Remediation Pathway

TAILINGS: PERMANENT LAND CONTAMINATION

Tailings are composed of salt, brine and clay. They are held in Tailings Management Areas (TMA)

GYPSTACKS: SEEPAGE RISK

Phosphogypsum (phosphorous fertilizer by-product)mounds contain naturally occurring radioactive material (NORM) and acidic leachate that threaten surrounding soil and groundwater.

23 MILLION M^3 INJECTED UNDERGROUND

Saline process water is permanently disposed into deep geological wells — an irreversible act with unknown long-term aquifer consequences.

29 MILLION TONNES OF WASTE — ANNUALLY

Non-hazardous mining by-products accumulate year over year, with surface disposal posing ongoing ecosystem exposure risk.

DEFERRED FINANCIAL LIABILITIES

$1.3B

ASSET RETIREMENT OBLIGATIONS

Legally required future reclamation costs

$343M

ACCRUED ENVIRONMENTAL COSTS

Contingent liabilities for active contamination remediation

COMBINED LIABILITY	$1.7B

Hot Spot 2 and 3 Fertilizer Runoff

System Condition 2 during Use

CHEMICAL CONCENTRATION



Planetary Boundaries Breached

N & P FLOWS — BOTH EXCEEDED

- Nitrogen and phosphorus biogeochemical flows rank among the most critically exceeded planetary bounds
- Fertilizer use is the primary driver of this issue, continuously pushing Earth's biogeochemical systems beyond their safe operating limits.

System Condition 3 during Use

ECOSYSTEM DESTRUCTION



Dead Zones & Algal Blooms

EUTROPHICATION POTENTIAL

- Nutrient runoff fuels explosive algal blooms that rapidly deplete dissolved oxygen in waterways, producing hypoxic dead zones in which aquatic life cannot survive.
- The resulting biodiversity collapse affects all living organism that inhabit the effected area

Future State Vision & UN SDG Alignment

OUR PROPOSED 2035 VISION

"Nutrien: the world's safest, lowest-carbon crop-input producer — closing the nutrient loop from mine to field, with zero preventable harm to workers or watersheds."



SDG 2 · Zero Hunger

TARGET 2.4

Sustainable, productive food systems — already Nutrien's stated ambition, so this is the goal we hold them to, not sell them.



SDG 6 · Clean Water

TARGET 6.3

Reduce untreated pollution released to water — our nutrient-runoff and tailings hotspots (SC2 + SC3).



SDG 8 · Decent Work

TARGET 8.8

Safe and secure working environments — our SC4 hotspot after three fatalities in 2024–25.



SDG 13 · Climate Action

TARGET 13.2

Integrate climate measures into strategy — our SC1 fossil-gas dependence in nitrogen production.

Each goal is anchored to a specific UN target and a live SLCA hotspot — chosen because our analysis found the gap, not because they photograph well.

Where Nutrien Sits Today — and the 2035 Target Phase

Sustainability Path	Rejection	Non-responsiveness	Compliance	Efficiency	Strategic Proactivity	The Sustaining Corporation
Human SC4 → SDG 8					TODAY	2035 TARGET
Economic cross-cutting → SDG 2				TODAY	2035 TARGET	
Environmental SC1–SC3 → SDG 6, 13				TODAY	2035 TARGET	

HUMAN

Strategic Proactivity → Sustaining Corporation

Governance already links safety, EDI and a disclosed Indigenous Strategy. The shift: workforce capability becomes the growth engine — Zero Harm (2027) plus re-skilling from 65%+ automated mining.

ECONOMIC

Efficiency → Strategic Proactivity

Today sustainability is argued as cost-and-returns. The shift: low-carbon ammonia from 1.2 Mt to 3+ Mt makes decarbonisation a revenue line, with GHG screens gating capex from 2028.

ENVIRONMENTAL

Efficiency → Strategic Proactivity

Headline results are intensity gains (−15% GHG/t) that volume growth can mask. The shift: absolute Scope 1+2 targets and CCUS at two more nitrogen sites change the asset base.

Why not The Sustaining Corporation on all three? A single-industry producer with a chemically emissions-intensive core product cannot credibly claim industry-agenda leadership by 2035 — Human is the one path where the capability and partnership foundations already exist.

Source: Perrott, B. (2014), “The sustainable organisation: blueprint for an integrated model,” Table I; Nutrien 2024 Global Sustainability Report.

Backcasting: Guiding Questions & Top Priorities

THE THREE GUIDING QUESTIONS

1) Does it move Nutrien in the right direction on all four system conditions? 2) Does it provide a flexible platform for future improvement? 3) Is it likely to produce sufficient return to fund the next step?

1

Low-Carbon Ammonia & CCUS Scale-Up

Directly attacks the largest red hotspot (SC1/SC2); existing Redwater/Alberta Carbon Trunk Line pathway proves ROI; opens access to green-premium and carbon-linked financing.

2

Zero Harm Safety & Automation Rollout

Responds to the most urgent hotspot (SC4) after 3 fatalities; automation already cuts both injury exposure and cost self-funding over time.

3

Precision Ag & Enhanced-Efficiency Fertilizer Push

Cuts runoff (SC2) at the point of use; monetizable through the existing 1,800+ location retail network and premium product margins.

4

Tailings & Phosphogypsum Circularity Program

Targets the SC3 land hotspot; reduces long-term reclamation liability and future regulatory/legal exposure.

Brainstorm: 13 Ideas Across Four Hotspots

Unfiltered and unranked — generated against the four SLCA hotspots before any prioritisation.

Climate & Emissions · SC1 + SC2

- Scale green/blue ammonia using renewable H_2 and CCUS
- Expand CO_2 capture (Alberta Carbon Trunk Line model) to more nitrogen sites
- Set Science Based Targets (SBTi) beyond the internal 2030 commitment

Land & Circularity · SC3

- Accelerate automated / tele-remote mining across all potash and phosphate sites
- Scale ultrafine-flotation tailings-reduction technology company-wide
- Develop phosphogypsum beneficial-reuse pathways (construction, soil amendments)

Nutrient Loss & Growers · SC2 + SC3

- Expand enhanced-efficiency fertilizers (controlled-release, nitrification inhibitors)
- Grow precision-agriculture / variable-rate tools via the retail network
- Launch grower watershed partnerships (buffer strips, cover-crop incentives)
- Expand the grower carbon program (insets/offsets) to more acres

People & Safety · SC4

- Launch a company-wide “Zero Harm” safety transformation with independent audits
- Extend Indigenous procurement / equity models beyond potash to all segments
- Publish site-level safety leading indicators, not just lagging fatality counts

Next: the three backcasting questions filter these 13 down to the four we fund.

From Baseline to Backcasted 2035

Executive Transition Architecture: Bridging Operational Risk to Long-Term Enterprise Resilience

MSCI AA RATING | VISION 2035

01 / CURRENT BASELINE

4 Material Hotspots

Core segment-level operational exposures

- Fossil-Dependent Nitrogen
- Land & Tailings Impacts
- Worker Safety Hazards
- Nutrient Runoff

Immediate Risk Containment Focus

02 / THE STRATEGIC BRIDGE

4 Sequenced Initiatives

Translating ambition into a fundable, board-ready plan

- Decarbonization & Low-Carbon Amonia →
- Tailings & Circularity →
- Zero-Harm Safety & Automation →
- Precision Agriculture & Runnoff reduction →

Phased Capital Allocation & ROI

03 / TARGET HORIZON

2035 Backcasted Vision

Aligned to 4 United Nations SDGs

- SDG 2 Zero Hunger
- SDG 6 Clean Water
- SDG 8 Decent Work
- SDG 13 Climate Action

Sustainable Moat & ESG Leadership

TEAM STRATEGIC TAKEAWAY

> *"Sustainability and competitiveness are the same project here every fix we prioritized also protects margin, capital access, or license to operate."*

MARGIN | CAPITAL ACCESS | LICENSE TO OPERATE

Pressure-Testing the TNS Strategy Through Four Lenses

1 Financial Lens

What is required to create value?

- MSCI AA rating vs. segment-level hotspots — a valuation gap peers can exploit
- Automation and efficiency gains are self-funding; protect ROIC
- Carbon pricing erodes the momentum case at gas-intensive nitrogen sites

2 Market Lens

Are these profitable, growing markets?

- ~10 billion people to feed by 2050 sustains core crop-input demand
- Low-carbon ammonia and enhanced-efficiency fertilizer are premium adjacencies
- Runoff and carbon regulation across NA/EU is reshaping the profit pool

Value-Creating Strategy Choices

Nutrien's four TNS priority projects, 2027–2035

3 Competitive-Advantage Lens

What does it take to win here?

- Largest upstream producer; majority of accessible global potash reserves
- Redwater / Alberta Carbon Trunk Line gives a CCUS head start peers lack
- 1,800+ retail locations — unmatched channel for precision ag

4 Operating-Model Lens

Do we have an organization that can deliver?

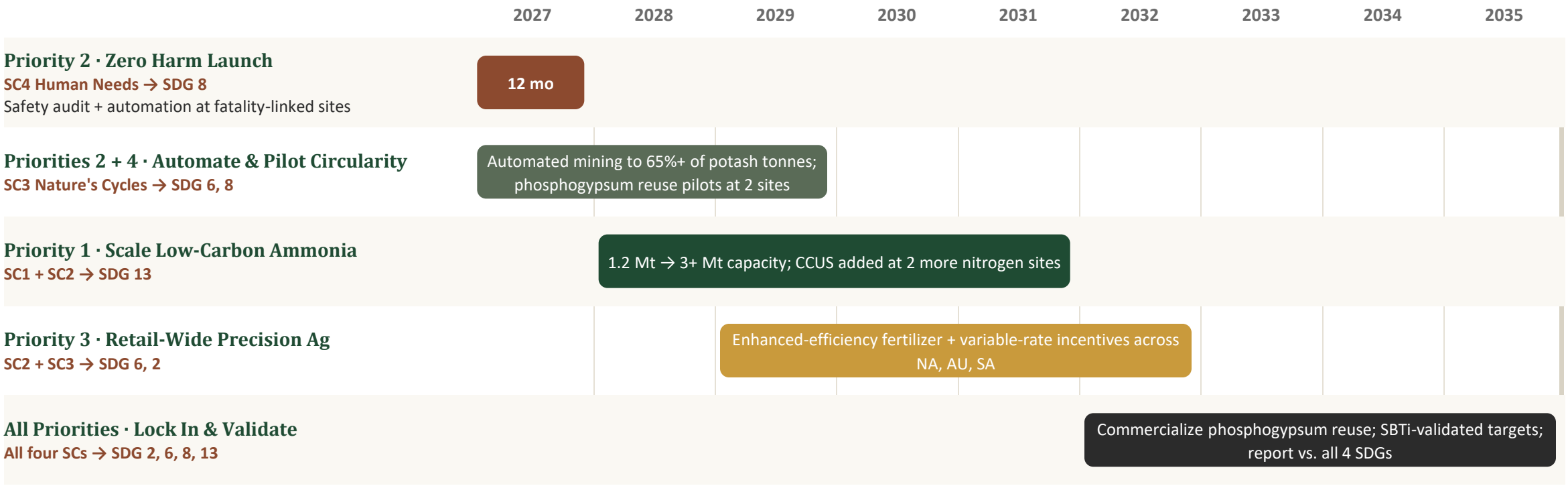
- Three 2024–25 fatalities expose a safety-management capability gap (SC4)
- One sequenced 2027–2035 capital program, not isolated initiatives
- Site-level leading indicators and SBTi validation to manage performance

All four lenses converge: the projects that close our TNS hotspots are the same projects that create value sustainability and competitiveness are one strategy here.

Source: McKinsey on Finance No. 68, “Seeing your way to better strategy” (2018)

Project Plan: Sequenced Roadmap to the 2035 Vision

A = Awareness (done) → B = Baseline (our SLCA + Perrott matrix) → C = Compelling vision (2035) → D = Down to action, below. Funded as one sequenced capital program.



Bar length = duration, 2027–2035 · Colour = TNS hotspot owner: rust = SC4 safety · sage = SC3 land · green = SC1/SC2 climate · gold = SC2/SC3 runoff · charcoal = all four

Each block also moves a Perrott path: Zero Harm and re-skilling drive Human toward The Sustaining Corporation; CCUS moves Environmental to Strategic Proactivity; retail incentives pull growers into shared plans, moving Economic; SBTi validation locks both in by 2035.

Sources: Nutrien 2024 Global Sustainability Report; Nutrien Form 40-F (FY2024); team backcasting analysis (ABCD — Down to Action).

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